

ITI Cmos Analog IC Design – Cadence Tools

Lab 01 Report — RC Low Pass Filter & MOSFET Characteristics

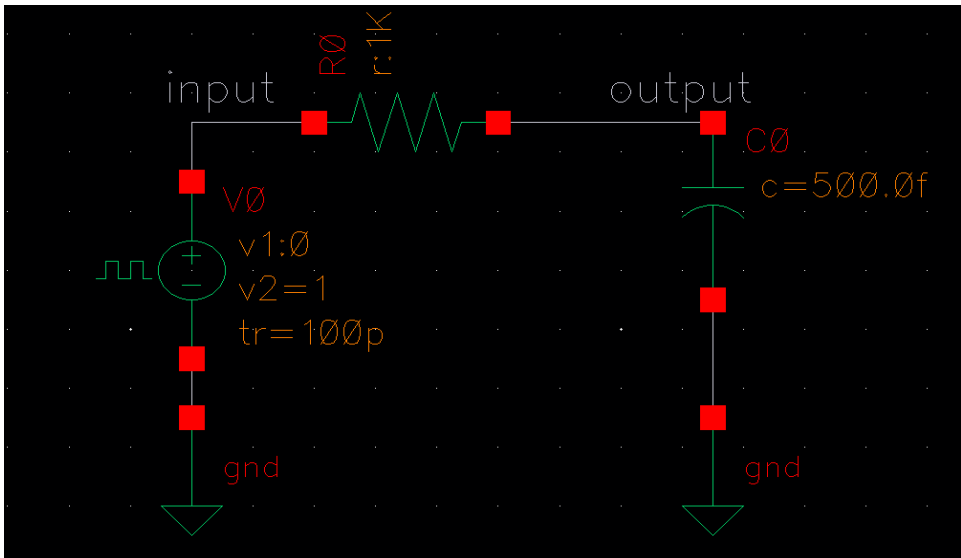
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Part 1: Low Pass Filter (LPF) Simulation

Assigned Parameters

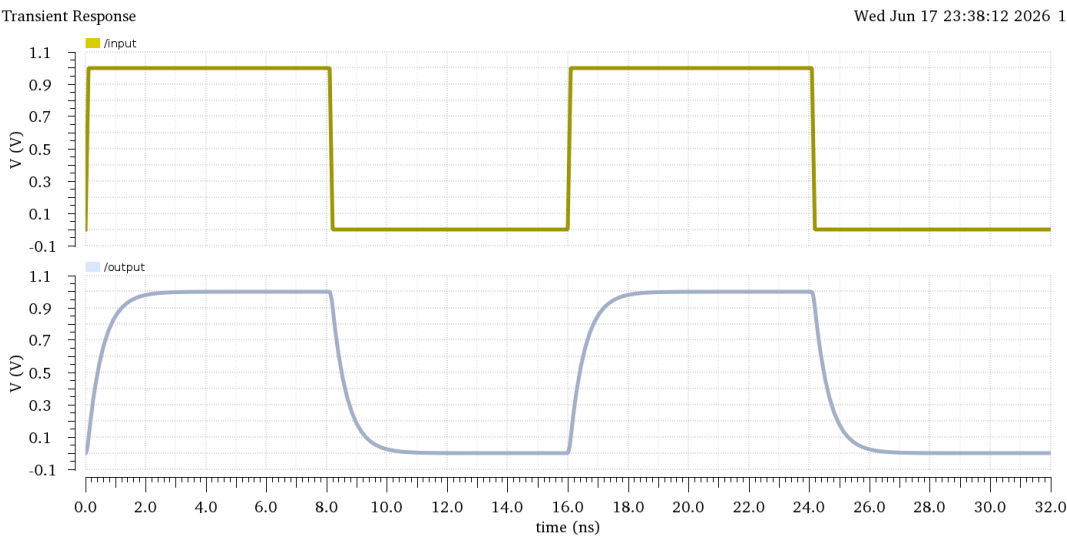
Parameter	Value	Notes
Time constant (τ)	0.5 ns	Given, fixed
Period (T)	16 ns	Randomly generated
R	1 k Ω	Given, fixed
C (calculated)	0.5 pF	$C = \tau / R = 0.5\text{ns} / 1\text{k}\Omega$

- schematic



1. Transient Analysis

1.1 Transient Results



1.2 Rise / Fall Time

labs:lpf:1	fall_time	1.095n			
labs:lpf:1	rise_time	1.098n			

Analytical rise/fall time for a first-order RC step response (10%–90%):

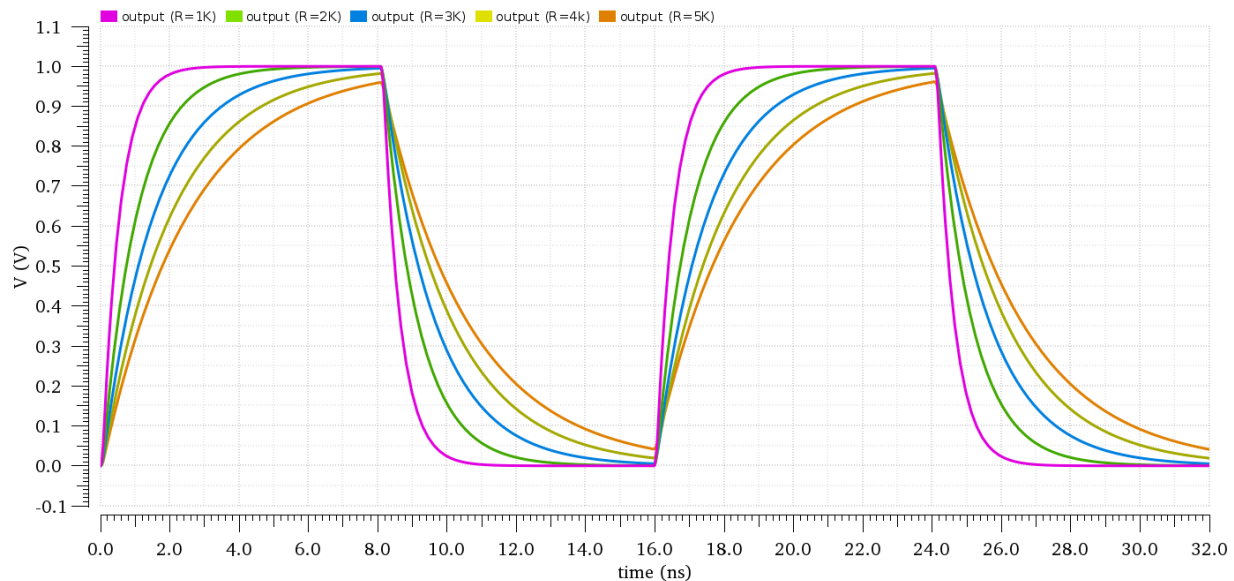
$$tr \approx tf \approx 2.2 \times \tau = 2.2 \times 0.5 \text{ ns} = 1.1 \text{ ns}$$

Parameter	Analytical	Simulated
fall time (10–90%)	1.1 ns	1.098 ns
Rise time (90–10%)	1.1 ns	1.095 ns

1.3 Parametric Sweep: R = 1kΩ to 5kΩ (step 1kΩ)

Transient Response

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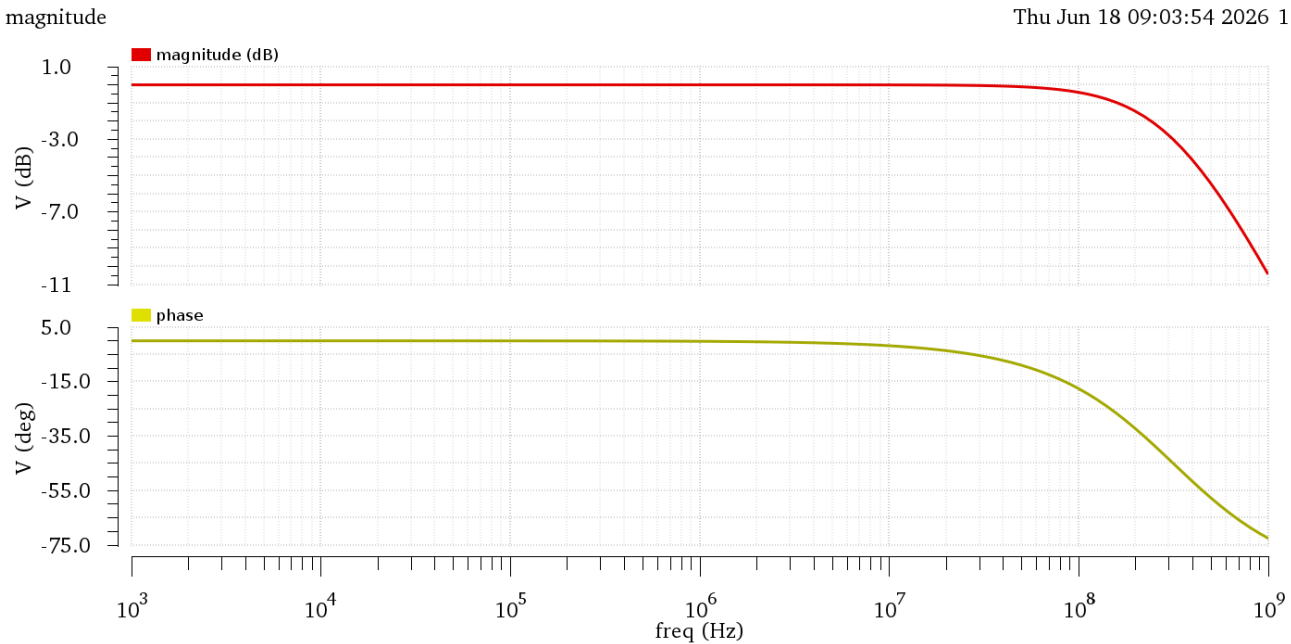


As r increase from 1k to 5k the rise time and fall time increase too, this happen because of time constant of the circuit as r increase and cap stay constant the time constant become bigger.

$$tr \approx tf \approx 2.2 \times \tau = 2.2 \times 0.5 \text{ ns} = 1.1 \text{ ns}$$

2. AC Analysis

2.1 Bode Plot (Magnitude & Phase)



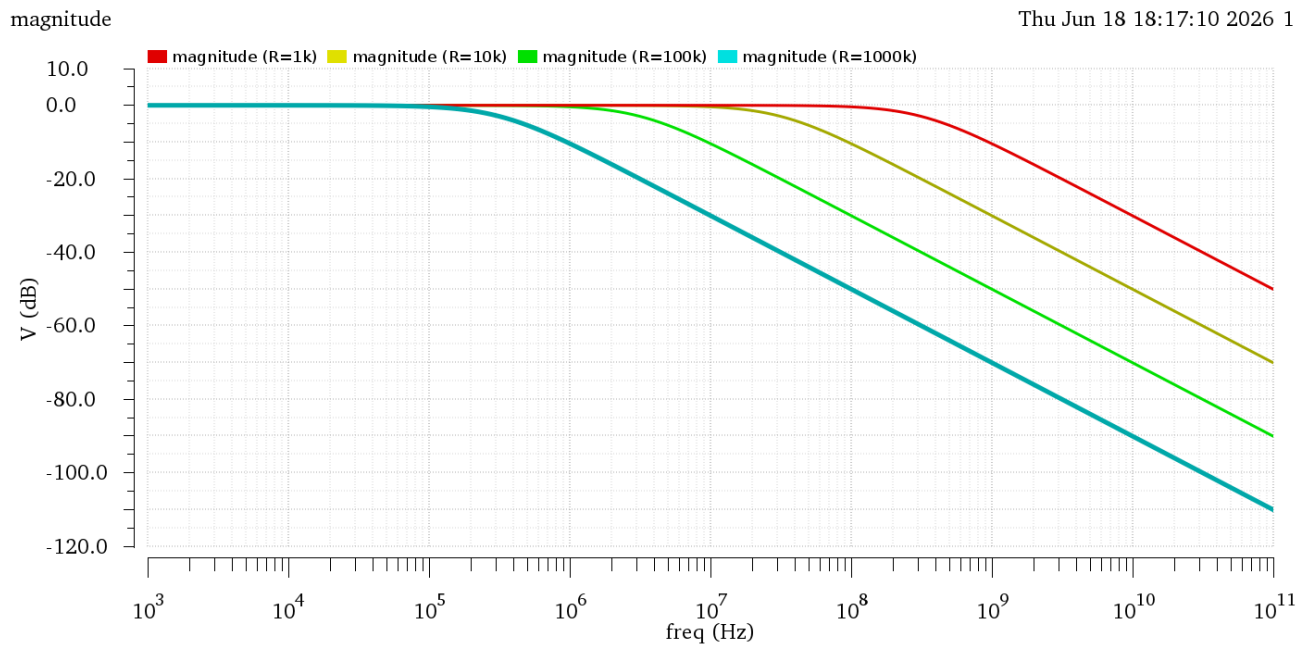
2.3 DC Gain & 3dB Bandwidth

labs:lpf:1	BW	317.6M			
labs:lpf:1	dc gain	0			

Analytical 3dB bandwidth: $f_{3dB} = 1 / (2\pi\tau) = 1 / (2\pi \times 0.5 \text{ ns}) \approx 318.3 \text{ MHz}$

Parameter	Analytical	Simulated
DC Gain	1 (0 dB)	1 (0 dB)
3dB Bandwidth	318.3 MHz	317.6 MHz

2.5 Parametric Sweep: R = 1, 10, 100, 1000 kΩ



As R increases, the 3dB bandwidth decreases proportionally ($f_{3dB} = 1/2\pi RC$), while the DC gain stays fixed at 0dB , but unity DC gain is preserved regardless of R.

3. [Optional] Pole-Zero Analysis

3.1 Pole-Zero Results

Poles (Hz)			
	Real	Imaginary	Qfactor
1	-3.18310e+08	0.00000e+00	5.00000e-01
No zero is found			

3.2 Pole Frequency vs. AC Bandwidth Comparison

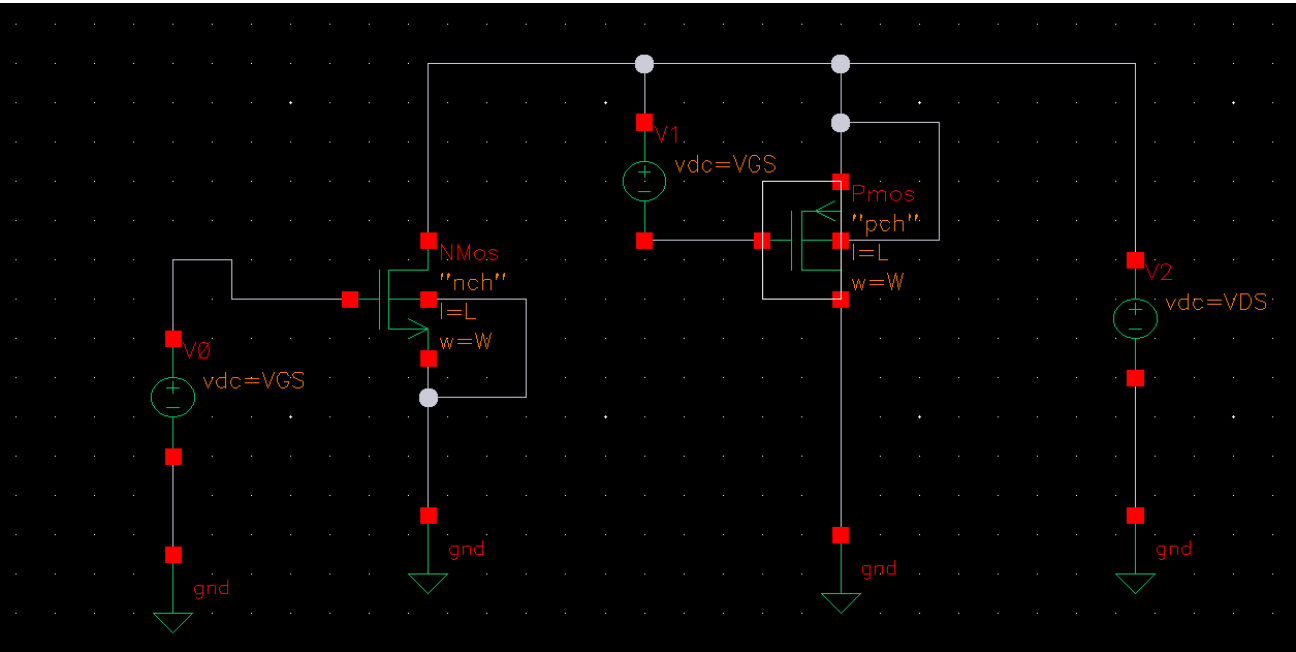
Analytical pole frequency: $f_{pole} = 1/(2\pi RC) = 1/(2\pi \times 0.5ns) \approx 318.3\text{ MHz}$

Analysis	Value
Pole-Zero analysis	318.3 MHz
AC analysis	317.6 MHz
Analytical	318.3 MHz

Part 2: MOSFET Characteristics

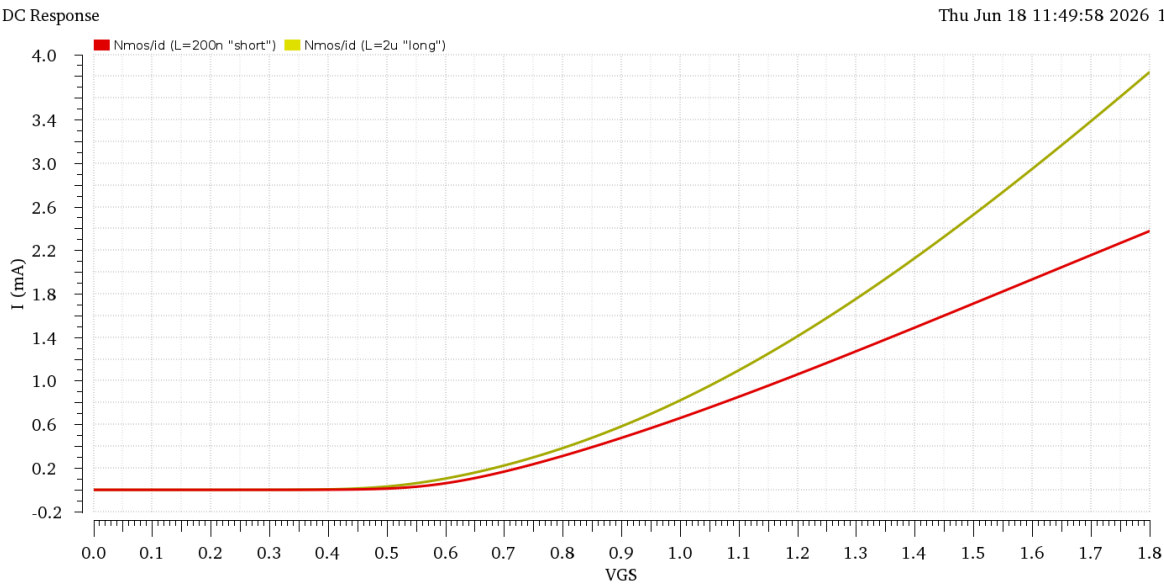
Device Sizes Used

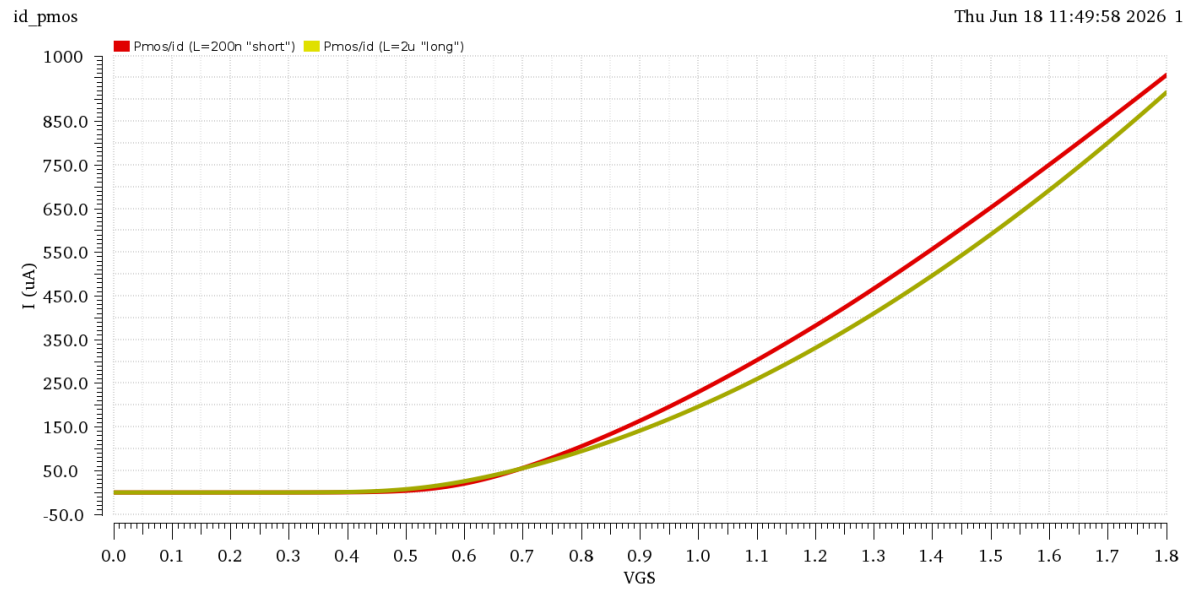
Device	W	L	W/L ratio
Short channel	4u	200n	20
Long channel	40u	2u	20



1. ID vs. VGS

1.1 ID-VGS Plots





1.2 Short Channel vs. Long Channel — Discussion

in NMOS long channel have high current while in NMOS short channel have high current
With identical W/L for both sizes, NMOS long channel since short-channel NMOS loses more current to velocity saturation, while PMOS favors short channel (since PMOS is less affected by velocity saturation).

The long channel device follows the quadratic relation square-law MOSFET behavior in saturation. The short channel device goes toward a more linear relation at higher V_{GS} , due to velocity saturation

1.3 NMOS vs. PMOS — Discussion

NMOS has higher current than PMOS at the same W/L and V_{GS} , because electron mobility is higher than hole mobility in silicon

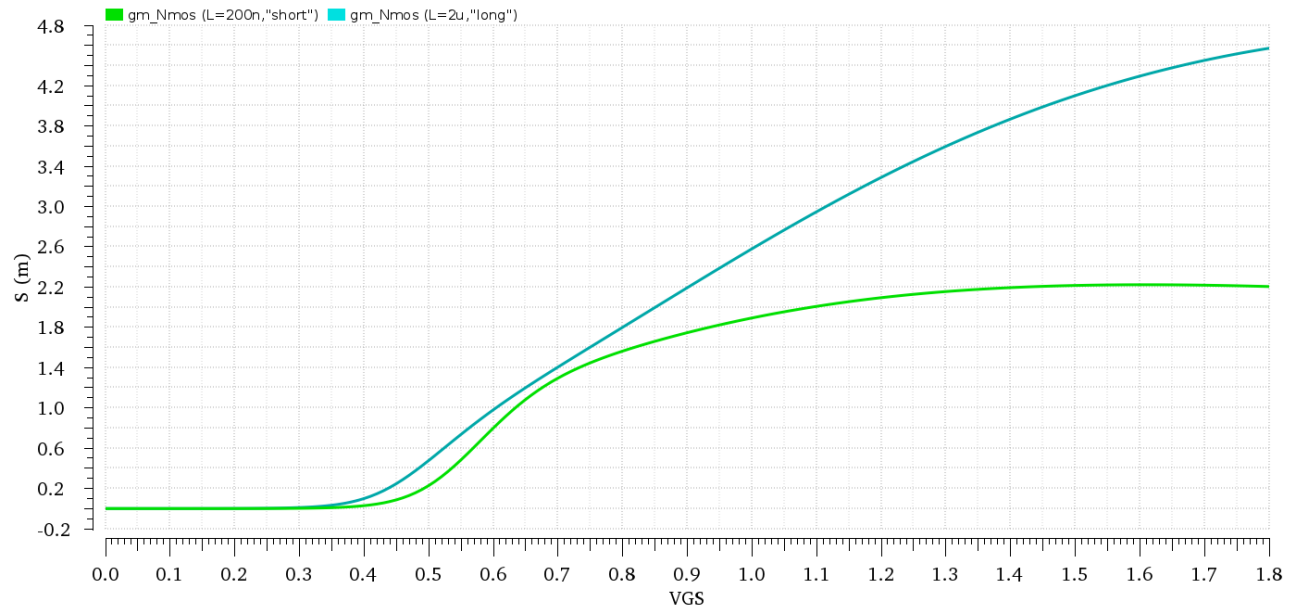
NMOS is typically more affected by short channel effects than PMOS, since electrons have higher mobility and saturation velocity effects tend to be more pronounced for NMOS devices at short channel lengths

2. gm vs. VGS

2.1 gm-VGS Plot

DC Analysis `dc`: VGS = (0 -> 1.8)

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2.2 Discussion

In Long channel gm increase linearly with VGS, following the ideal square-law.

In short channel gm increase slowly with VGS because of velocity saturation

In Long channel gm does not saturate it keeps rising since the long channel device follows square-law behavior

Short channel: gm saturates at high VGS, because of velocity saturation.

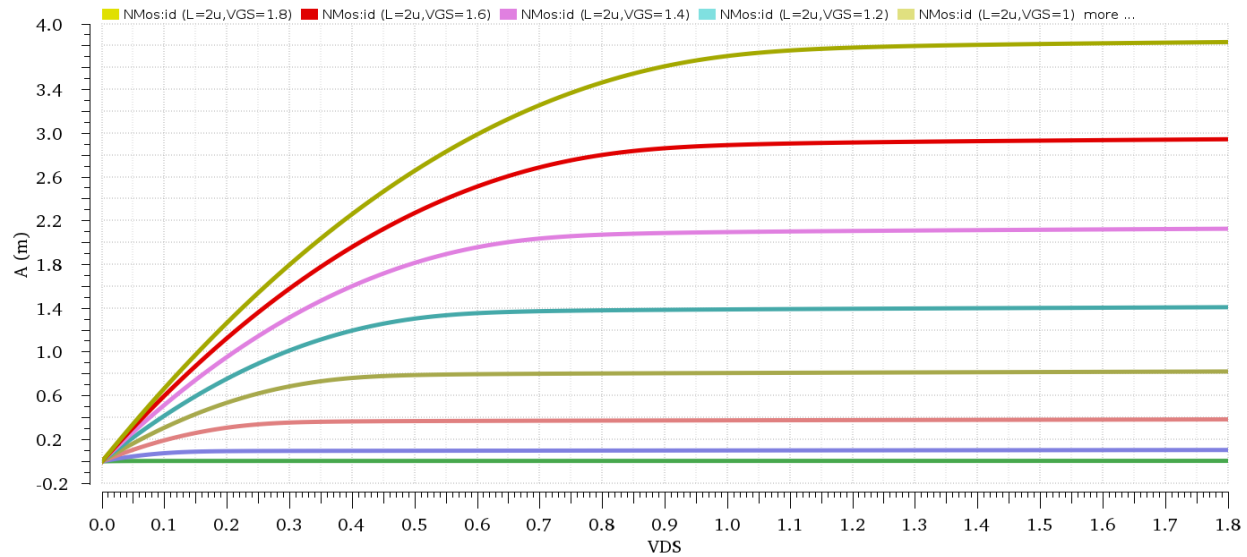
3. ID vs. VDS

3.1 ID-VDS

- Long channel

DC Analysis `dc`: VDS = (0 -> 1.8)

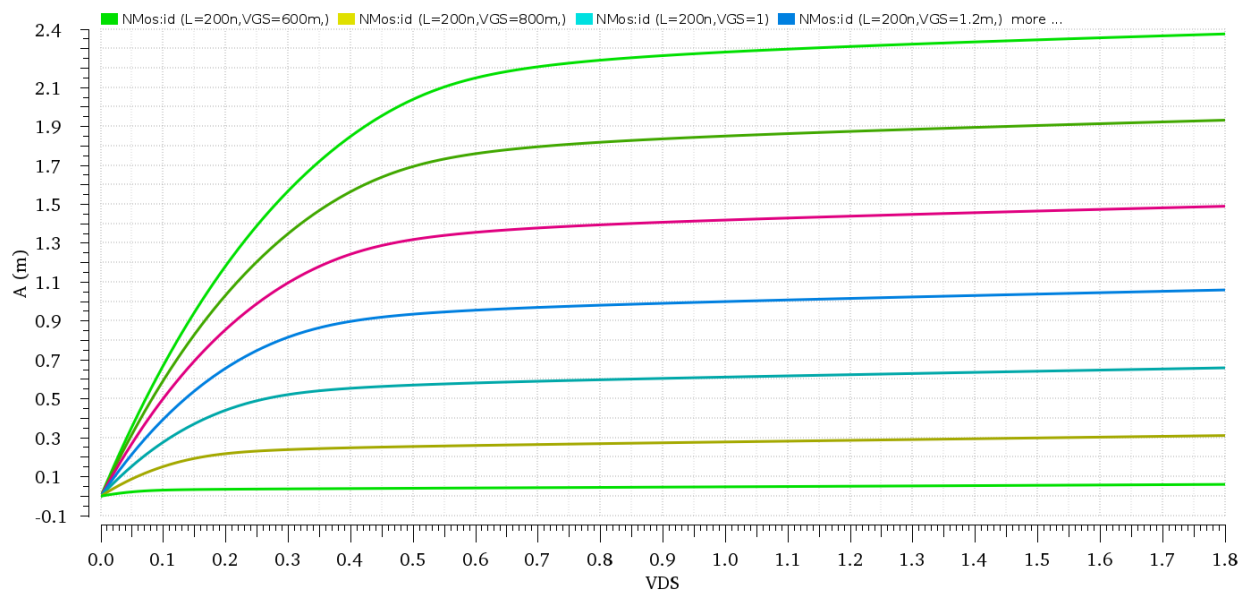
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- Short channel

DC Analysis `dc`: VDS = (0 -> 1.8)

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3.2 Discussion

Long channel device has higher current because the short channel device suffers from velocity saturation

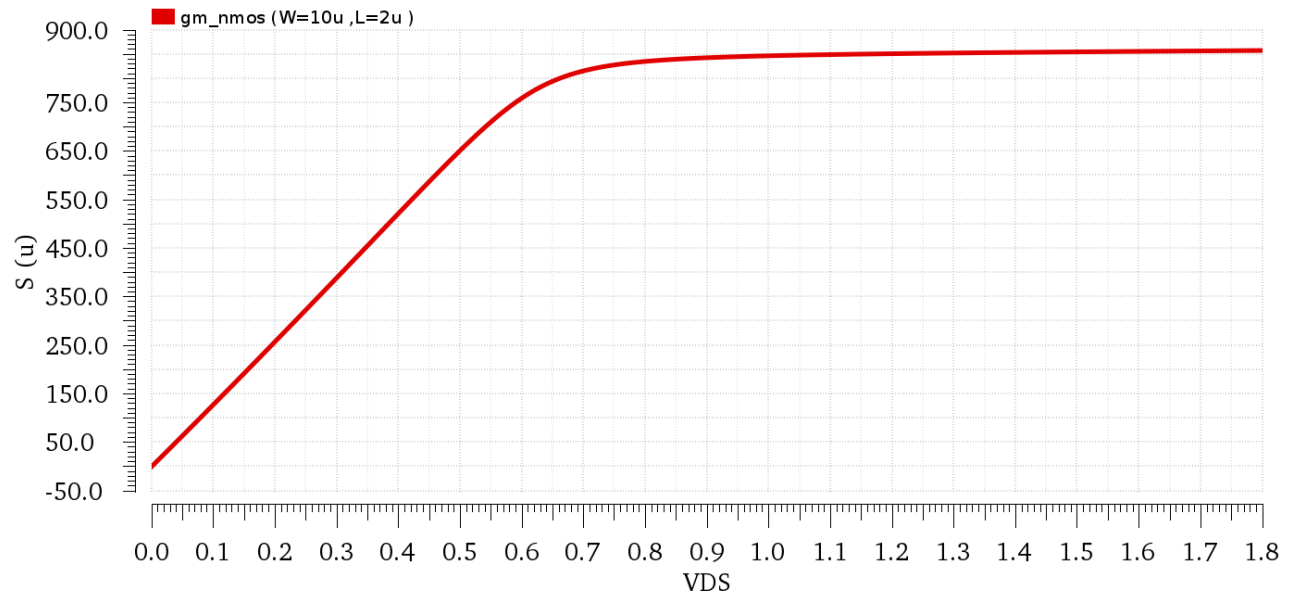
Short channel device has a higher slope because with short L channel length modulation is stronger and takes effect on current

4. [Optional] gm and ro in Triode and Saturation

4.1 gm vs VDS Plot

DC Response

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4.2 Discussion — gm

in the triode region g_m increases linearly with V_{DS} . In triode, I_D depends on both V_{GS} and V_{DS} based on square law equation

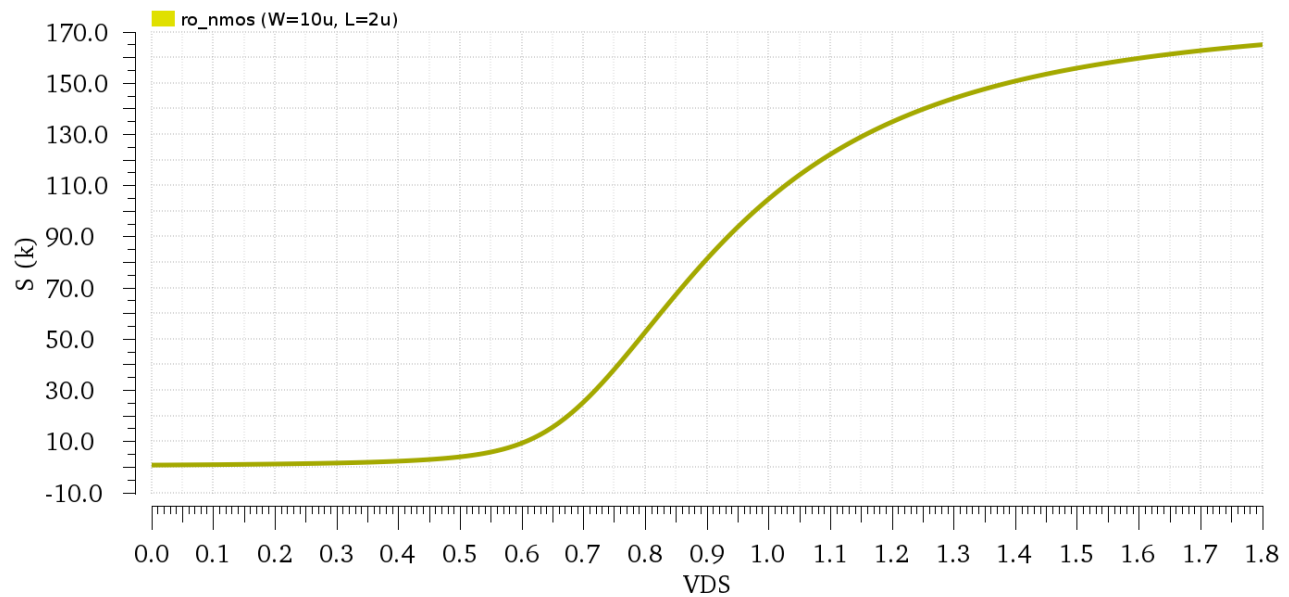
g_m saturates once the device enters saturation. In saturation, I_D no longer depends on V_{DS} in the square-law equation.

Deep in saturation of course g_m is at its maximum and constant value.

4.3 r_o vs V_{DS} plot

DC Response

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4.4 Discussion — r_o

r_o does not saturate after entering saturation region.

deep into saturation, the higher r_o becomes. After that it become to decrease again at high V_{DS}

the edge of saturation should be avoided for analog applications. At the edge, r_o is still low